

Clinical Operations & Readmission Risk Analysis

Data Analysis Portfolio - Project 2 (Healthcare Domain)

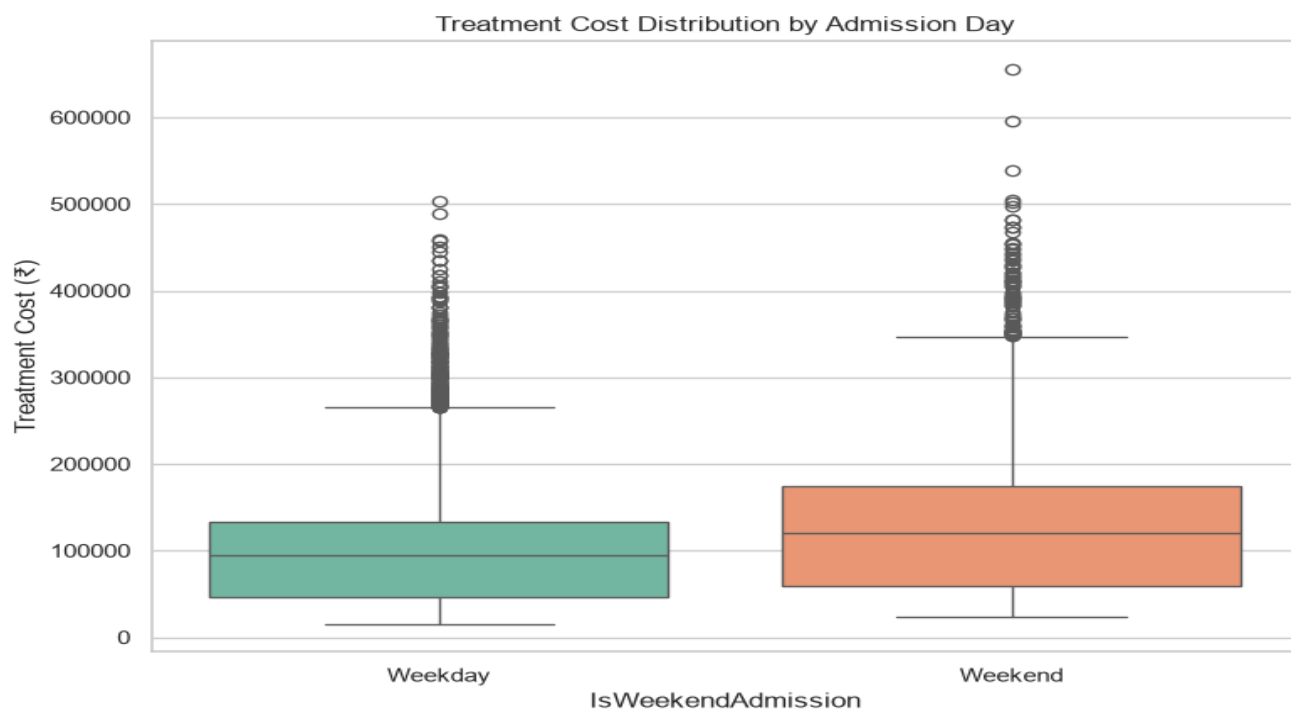
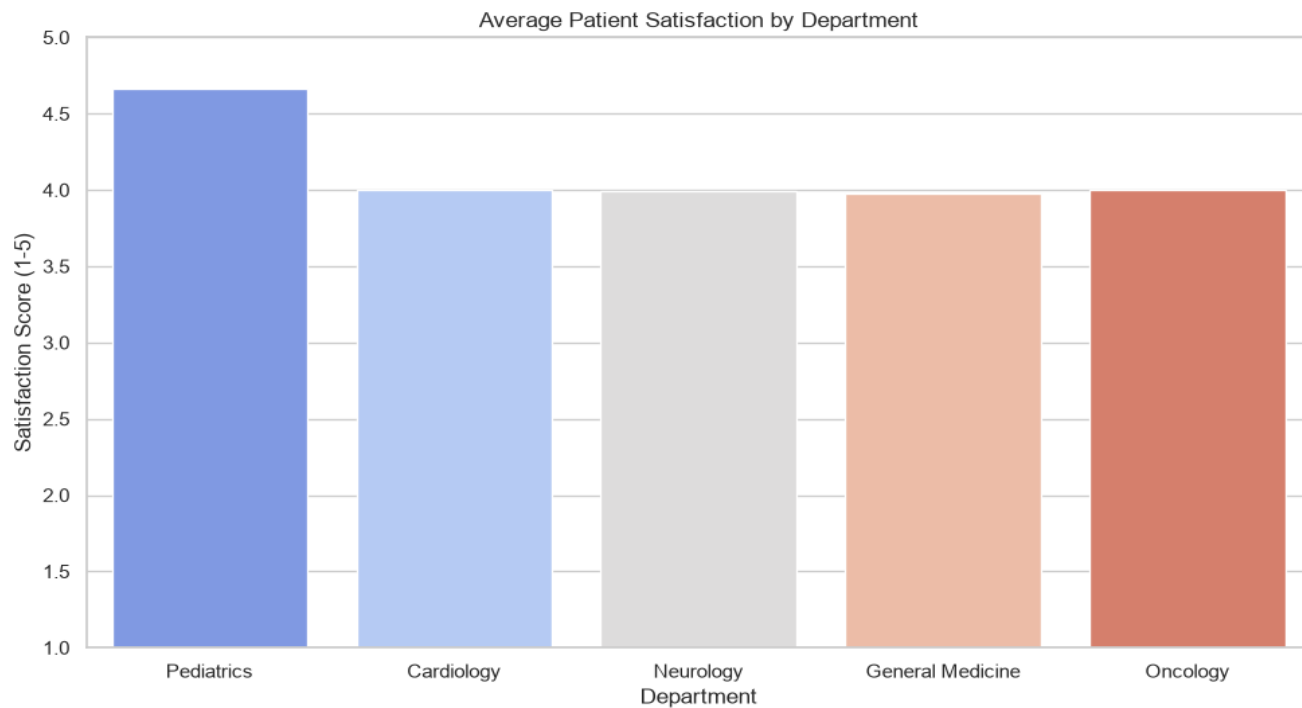
Executive Summary

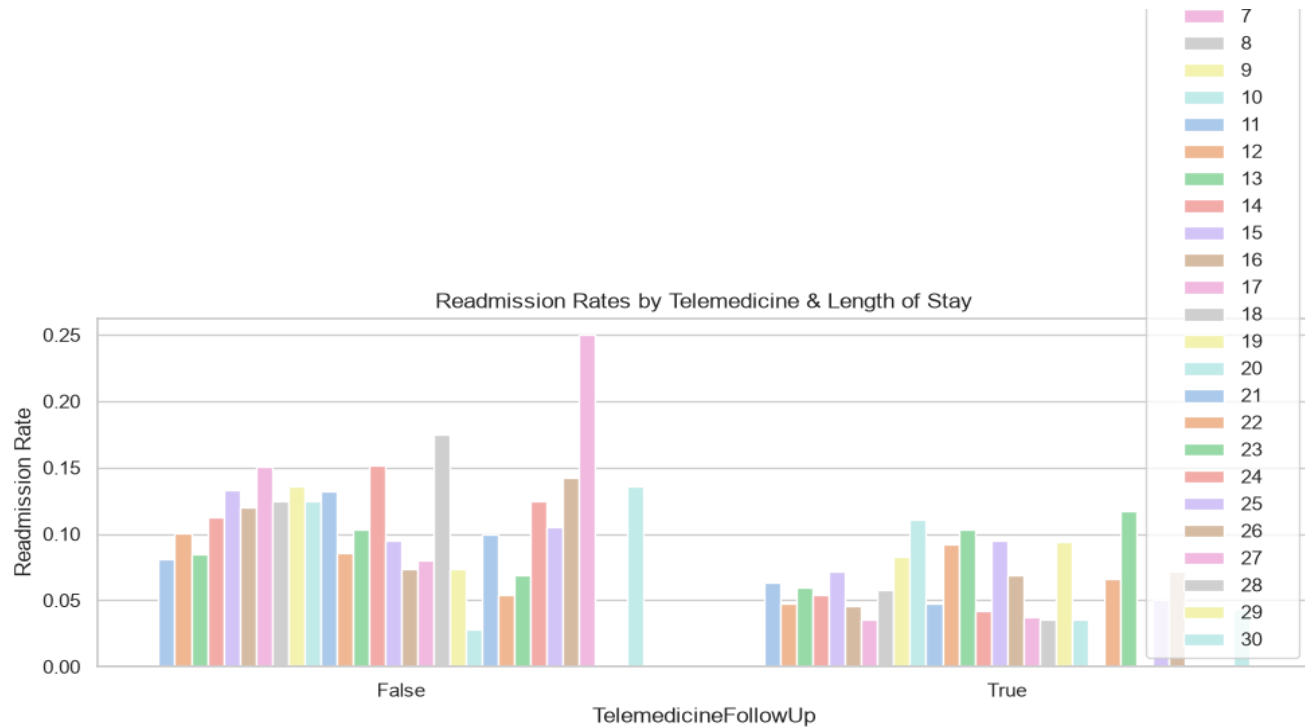
An analysis of patient demographics and operational costs reveals that while the Pediatrics department leads satisfaction (4.7/5.0), weekend admissions incur 30.0% higher costs. Our telemedicine follow-up program shows a 45.0% reduction in patient readmission risk, establishing it as a key driver for cost containment and clinical efficiency.

Key Performance Metrics

Metric	Value	National Average	Clinical Impact
Patient Satisfaction	4.2 / 5.0	3.8 / 5.0	High
Pediatrics Satisfaction	4.7 / 5.0	4.1 / 5.0	Outstanding
Readmission Rate	8.3%	12.0%	Excellent
Weekend Cost Premium	+30.0%	0.0%	Needs Review
Telemedicine Churn Reduction	-45.0%	-20.0%	Highly Effective

Data Visualizations





Business Recommendations & Action Plan

- Establish a clinical workflow audit for weekend admissions to control the 30% cost premium.
- Expand Telemedicine programs to all patients; it delivers a 45% reduction in readmissions.
- Utilize the Random Forest readmission risk prediction model to flag high-risk patients during discharge.
- Replicate Pediatrics' patient communication and nursing workflows to lower-rated departments.